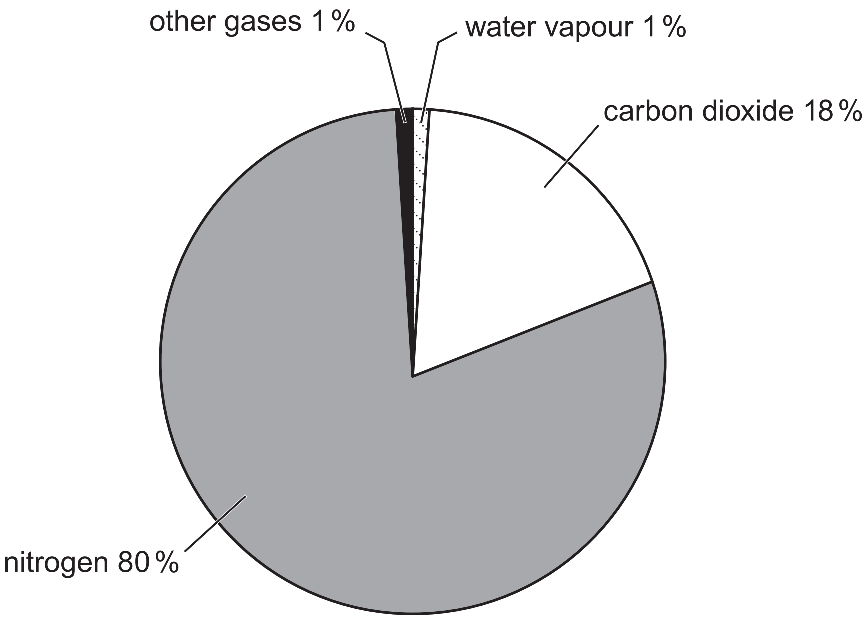
Examiner only

Answer **all** *questions.*

1. *(a)* The Earth's atmosphere has evolved over millions of years to its present composition.

The pie chart shows the composition of the atmosphere at one stage of its evolution.



(i) Describe the **main** differences in the composition of the atmosphere shown in the pie chart and that present on the Earth today. [3]

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(ii) **Complete the table** to describe simple chemical tests that can be used in the laboratory to identify hydrogen and carbon dioxide. [2]

Gas	Test carried out	Expected observation
hydrogen		
carbon dioxide		

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(b) During the last 250 years the level of carbon dioxide in the atmosphere has increased. Most scientists believe that the increase in the level of carbon dioxide in the atmosphere has resulted in global warming.

The table below shows some data about the atmosphere between 1750 and 2000.

Year	Concentration of carbon dioxide in the atmosphere (parts per million)	Average global temperature (°C)
1750	278	13.3
1800	282	13.4
1850	288	13.5
1900	297	13.7
1950	310	14.0
2000	368	14.6

Use the data in the table to answer parts (i) and (ii).

(i) Describe the trend in average global temperature between 1750 and 2000. [2]

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(ii) Calculate the percentage increase in the concentration of carbon dioxide in the atmosphere from 1750 to 1850. [2]

percentage increase = ... %

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